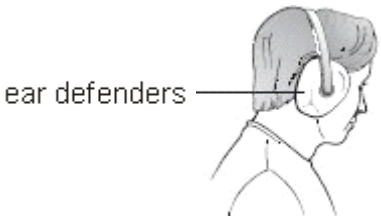
Q1. Three pupils watched a firework display.

(a) A man lit the fireworks. He wore ear defenders.



Why should he wear ear defenders when he is close to loud fireworks?

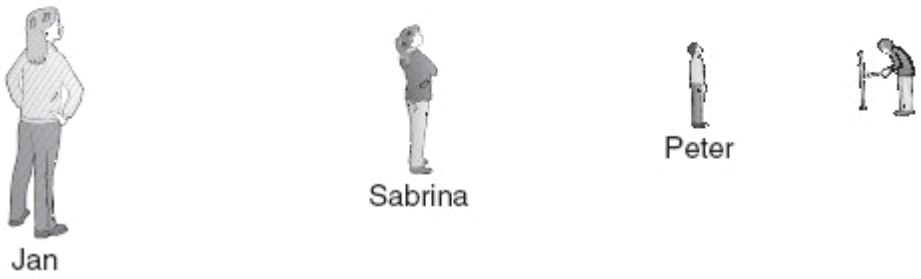
.....

.....

1 mark

(b) A rocket exploded making a loud sound and a bright flash.
Peter, Sabrina and Jan were standing at different distances from the rocket.





When the rocket exploded, Jan heard the quietest sound.
Why did Jan hear the quietest sound?

.....

.....

1 mark

(c) Jan saw the flash before she heard the sound.

What does this tell you about the speed of light and the speed of sound?

.....

.....

1 mark

(d) Complete the sentences below using words from the list.

chemical **electrical** **heat** **light** **sound**

(i) Jan, Sabrina and Peter could **see** the rocket explode because it gave out ..... energy.

1 mark

(ii) They could **hear** the rocket explode because it gave out ..... energy.

1 mark

(e) When the rocket stopped burning it fell to the ground.
What force caused it to fall to the ground?

.....

1 mark

maximum 6 marks

Mark schemes

Q1.

(a) any **one** from

- loud sounds can damage hearing
  accept 'they could damage his hearing'
- to protect his ears
  accept 'loud sounds can damage the ear'
- loud sounds can burst the eardrum
  accept 'it could make him deaf'
  accept 'they are very loud'
  'they are loud' is insufficient as 'loud' is given in the question

1 (L3)

(b) any **one** from

- the further the sound travels the quieter it gets
  *accept 'the sound **or** energy spreads out'*
- she was furthest away
  *accept 'she was further away' **or** 'she was far away'*
  accept 'some of the sound is absorbed'

1 (L3)

(c) any **one** from

- the speed of light is greater than the speed of sound
  accept 'sound travels more slowly'
- light travels faster than sound
  accept 'light travels faster'
  accept 'light gets there before sound'

1 (L4)

(d) (i) • light

1 (L3)

(ii) • sound

1 (L3)

(e) gravity
accept 'weight'

1 (L4)

[6]